Exercise 12.2 SOLUTION

Q1. **Determine the left hand limit and the right hand limit and then, find limit of the following functions when $x \to c$.**

(i) $f(x) = 2x^2 + x - 5$, $c = 1$

L.H.L $= \lim_{x\to 1^-} f(x) = \lim_{x\to 1} (2x^2+x-5)$

$= 2(1)^2 + 1 - 5 = 2+1-5 = -2$

R.H.L $= \lim_{x\to 1^+} f(x) = \lim_{x\to 1} (2x^2+x-5)$

$= 2(1)^2+1-5 = 2+1-5 = -2$

L.H.L = R.H.L = -2 so $\lim_{x\to 1} f(x) = -2$

(ii) $f(x) = \dfrac{x^2-9}{x-3}, c = -3$

L.H.L $= \lim_{x\to -3^-} f(x) = \lim_{x\to -3} \dfrac{x^2-9}{x-3} = \dfrac{(-3)^2-9}{-3-3} = \dfrac{9-9}{-6} = 0$

R.H.L $= \lim_{x\to -3^+} f(x) = \lim_{x\to -3} \dfrac{x^2-9}{x-3}$

$= \dfrac{(-3)^2-9}{-3-3} = \dfrac{9-9}{-6} = 0$

L.H.L = R.H.L = 0 So, $\lim_{x\to 1} f(x) = 0$

(iii) $f(x) = |x-5|$, $c = 5$

L.H.L $= \lim_{x\to 5^-} f(x) = \lim_{x\to 5} -(x-5) = -(5-5) = 0$

R.H.L $= \lim_{x\to 5^+} f(x) = \lim_{x\to 5} (x-5) = 5-5 = 0$

L.H.L = R.H.L = 0 So, $\lim_{x\to 5} f(x) = 0$

Q2. **Discuss the continuity of f(x) at $x = c$**

(i) $f(x) = \begin{cases} 2x+5 & \text{if } x \le 2 \\ 4x+1 & \text{if } x > 2 \end{cases}$; $c = 2$

L.H.L $= \lim_{x\to 2^-} f(x) = \lim_{x\to 2} (2x+5)$

$= 2(2) + 5 = 4 + 5 = 9$

R.H.L $= \lim_{x\to 2^+} f(x) = \lim_{x\to 2} (4x+1) = 4(2) + 1 = 9$

f(2) $= 2(2) + 5 = 4+5 = 9$

L.H.L = R.H.L = f(2)

so f(x) is Continuous at $x = 2$

(ii) $f(x) = \begin{cases} 3x-1 & \text{if } x < 1 \\ 4 & \text{if } x = 1 \\ 2x & \text{if } x > 1 \end{cases}$; $c = 1$

L.H.L $= \lim_{x\to 1^-} f(x) = \lim_{x\to 1} (3x-1)$

$= 3(1) - 1 = 3 - 1 = 2$

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R.H.L $= \lim_{x\to 1^+} f(x) = \lim_{x\to 1}(2x) = 2(1) = 2$

$f(1) = 4$

$f(1) \neq$ L.H.L = R. H. L

so $f(x)$ is Discontinuous at $x = 1$

Q3. If $f(x) = \begin{cases} 3x & \text{if } x \le -2 \\ x^2 - 1 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \ge 2 \end{cases}$,

Discuss continuity at $x = 2$ and $x = -2$

Case I for $x = 2$

L.H.L $= \lim_{x\to 2^-} f(x) = \lim_{x\to 2} (x^2-1) = (2)^2 - 1 = 4 - 1 = 3$

R.H.L $= \lim_{x\to 2^+} f(x) = \lim_{x\to 2} 3 = 3$, $f(2) = 3$

L.H.L = R.H.L = $f(2)$

$f(x)$ is Continuous at $c = 2$

Case II For $x = -2$

L.H.L $= \lim_{x\to -2^-} f(x) = \lim_{x\to -2^-} (3x) = 3(-2) = -6$

R.H.L $= \lim_{x\to -2^+} f(x) = \lim_{x\to -2^+} (x^2 - 1)$

$= (-2)^2 - 1 = 4 - 1 = 3$

$f(-2) = 3(-2) = -6$

L.H.L = $f(-2) \neq$ R.H.L

$f(x)$ is Discontinuous at $x = -2$

Q4. $f(x) = \begin{cases} x + 2 , & x \le -1 \\ c + 2 , & x > -1 \end{cases}$ **find c so that $\lim_{x\to -1} f(x)$ exist**

L.H.L $= \lim_{x\to -1^-} f(x) = \lim_{x\to -1}(x+2) = -1 + 2 = 1$

R.H.L $= \lim_{x\to -1^+} f(x) = \lim_{x\to -1}(c+2) = c + 2$

Given limit exist means; L.H.L = R.H.L

$1 = c + 2 \Rightarrow$ $\boxed{c = -1}$

Q5. **Find the values m and n, so that given function f is continuous at $x = 3$**

(i) $f(x) = \begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x + 9 & \text{if } x > 3 \end{cases}$

L.H.L $= \lim_{x\to 3} f(x) = \lim_{x\to 3} (mx) = 3m$

R.H.L $= \lim_{x\to 3^+} f(x) = \lim_{x\to 3}(-2x + 9)$

$= -2(3) + 9 = -6 + 9 = 3$ and $f(3) = n$

Given $f(x)$ is Continuous

So L.H.L = R.H.L = $f(3)$ $\Rightarrow$ $3m = 3 = n$

$\Rightarrow 3m = 3$ and $n = 3$

$\Rightarrow \boxed{m = 1}$ and $\boxed{n = 3}$

(ii) $f(x) = \begin{cases} mx & \text{if } x < 3 \\ x^2 & \text{if } x \ge 3 \end{cases}$

L.H.L $= \lim_{x\to 3^-} f(x) = \lim_{x\to 3} (mx) = 3m$

R.H.L $= \lim_{x\to 3^+} f(x) = \lim_{x\to 3} x^2 = (3)^2 = 9$

$f(3) = (3)^2 = 9$

Given $f(x)$ continuous mean

L.H.L = R.H.L = $f(3)$

$\Rightarrow 3m = 9 = 9 \Rightarrow \boxed{m = 3}$

Q6. $f(x) = \begin{cases} \dfrac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} & x \neq 2 \\ k & x = 2 \end{cases}$,

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Find value of K so that f is continuous at $x = 2$

$$\lim_{x\to 2} f(x) = \lim_{x\to 2} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} \times \frac{\sqrt{2x+5} + \sqrt{x+7}}{\sqrt{2x+5} + \sqrt{x+7}}$$

$$= \lim_{x\to 2} \frac{\left(\sqrt{2x+5}\right)^2 - \left(\sqrt{x+7}\right)^2}{(x-2)\left(\sqrt{2x+5} + \sqrt{x+7}\right)}$$

$$= \lim_{x\to 2} \frac{2x + 5 - x - 7}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x\to 2} \frac{\cancel{(x-2)}}{\cancel{(x-2)}\left(\sqrt{2x+5} + \sqrt{x+7}\right)} = \lim_{x\to 2} \frac{1}{\left(\sqrt{2x+5} + \sqrt{x+7}\right)}$$

$$= \frac{1}{\sqrt{2(2)+5} + \sqrt{2+7}} = \frac{1}{\sqrt{9} + \sqrt{9}} = \frac{1}{3+3} = \frac{1}{6}$$

Given $f(x)$ is continuous so

$\lim_{x\to 2} f(x) = f(2) \Rightarrow \frac{1}{6} = k$ *or* $\boxed{k = \frac{1}{6}}$

Q.7 **Given the function** $f(x) = \begin{cases} 2x + 3 , & x \le 1 \\ -x + 4 , & x > 1 \end{cases}$

Discuss the limit and continuity at $x = 1$

Sol: $f(1) = 2(1) + 3 = 2 + 3 = 5$

L.H.L $= \lim_{x\to 1^-} f(x) = \lim_{x\to 1}(2x + 3) = 2(1) + 3 = 2 + 3 = 5$

R.H.L $\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+}(-x + 4) = -1 + 4 = 3$

Since $\lim_{x\to 1} f(x) \neq \lim_{x\to 1^+} f(x)$

Limit does not exist at $x = 1$

Also f(x) is not continuous at x = 1

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